

**An Uncertainty Analysis for Carbon Quantification from Above-Ground Tree
Biomass: Predicting Uncertainties in Allometric Models in Canada and
Sweden**

By

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DECLARATION OF CO-AUTHORSHIP / PREVIOUS PUBLICATION

I. Co-Authorship

I hereby declare that this thesis incorporates material that is result of joint research. This thesis was completed under the supervision of Dr. Rajeev Ruparathna and Dr. Jerald Lalman from the University of Windsor (Windsor Ontario, Canada) and Dr. Shareq Mohd Nazir from the KTH Royal Institute of Technology (Stockholm, Sweden). In all cases, the key ideas, primary contributions, designs, data analysis, interpretation, and writing were performed by the author. The contributions of the co-authors were primarily provision of the research ideas, feedback throughout the experimental design, review of the results, participation in discussions, and the refinement and editing of the manuscript.

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Thesis Chapter	Publication Title/Full Citation	Publication Status
<i>Chapter [3]</i>	Simulation-Based Uncertainty Analysis Methodology for Carbon Quantification from Above-Ground Tree Biomass Allometric Models in Canada and Sweden	<i>In Preparation</i>

<i>Chapter [4]</i>	Evaluating the Effects of Model Uncertainties on Carbon Quantified from Above-Ground Tree Biomass Models: Adjustments and implications for methodological and political decision-making	<i>In Preparation</i>
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ABSTRACT

In compliance with international commitments to address the increasingly urgent need to reduce greenhouse gas (GHG) emissions, countries prepare national GHG inventories (NGHGI). NGHGIs include annual estimates of anthropogenic GHG emissions and removals. Reliable data in NGHGIs are essential for creating effective climate change policies and mitigation strategies, determining compliance with internationally agreed-upon targets, and tracking the sources and trends of GHG emissions and reductions. The above-ground biomass (AGB) carbon pool from the forestry sector is expected to contribute largely to carbon reductions; however, data for this sector is highly uncertain due to quantification challenges. These uncertainties have adversely impacted the reliability of the climate mitigation strategies and policies based on this data. AGB is quantified mainly by employing a Tier 3 approach involving allometric models derived from forest inventory data. A review of published literature indicated a need for research on the methods used to quantify model uncertainties and the effects of these uncertainties on carbon estimates from AGB. This research employs a simulation-based uncertainty analysis to quantify model uncertainties in carbon estimates from AGB allometric models. The literature, manuals, and R software were used to develop the uncertainty analysis. Alternative uncertainty analysis approaches were proposed to determine their effects on the uncertainty estimates. Case studies were performed for study areas in Canada and Sweden to determine the feasibility of the uncertainty analysis method when used in different countries. The results of this study demonstrated how model uncertainties can be quantified using the proposed uncertainty analysis method, and how estimates can be adjusted for uncertainties. The uncertainty estimates did not differ significantly from using the alternative uncertainty analysis methods. The main causes of model uncertainty for both case studies were due to measurement uncertainty in the model input variables and residual uncertainty. Recommendations were made on how uncertainties can be reduced by prioritizing methodological and data collection improvements in these areas. The effects of uncertainties on climate change mitigation strategies and methods to incorporate uncertainty information into climate change policies were assessed.

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LIST OF ABBREVIATIONS

AGB	:	Above-Ground Biomass
ArcGIS	:	Aeronautical Reconnaissance Coverage Geographic Information System
B.C.	:	British Columbia
CO ₂	:	Carbon Dioxide
COP	:	Conference of Parties
Dbh	:	Diameter at Breast Height
FLNRO	:	Forest, Lands, and Natural Resource Operations
GHG	:	Greenhouse Gas
Ht	:	Canopy Height
IPCC	:	Intergovernmental Panel on Climate Change
LULUCF	:	Land-Use, Land-Use Change, and Forestry
MCS	:	Monte Carlo Simulation
NDC	:	Nationally Determined Contribution
NGHGI	:	National Greenhouse Gas Inventory
PDF	:	Probability Density Function
SLU	:	Swedish University of Agricultural Sciences
TACCC	:	Transparency, Accuracy, Completeness, Comparability, Consistency
UN	:	United Nations
UNFCCC	:	United Nations Framework Convention on Climate Change
VRI	:	Vegetation Resource Inventory
WMO	:	World Meteorological Organization

1.0 INTRODUCTION

1.1 Background and Problem Statement

Despite accords established at the United Nations (UN) Conference of Parties (COP), commitments to reduce greenhouse gas (GHG) emissions and stabilize global temperature rise have not been on track to limit global warming (COP26 Explained, 2021; UNEP, 2022). The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment report emphasized the need for a rapid decline in GHG emissions, as climate resilient developments will become increasingly limited if global warming levels exceed 1.5°C (IPCC, 2022). Many countries acknowledged the urgency for climate action through commitments to achieve net-zero emissions within the next few decades (van Soest et al., 2021). For example, the Canadian government plans to transition to net-zero GHG emissions by 2050, and the Swedish government by 2045 (Canadian Institute for Climate Choices, 2021; Government Offices of Sweden Ministry of the Environment and Energy, 2018). However, policymakers are concerned with the feasibility of planning for net-zero amidst uncertainties in GHG emission and removal data (Canadian Institute for Climate Choices, 2021).

In compliance with the UN Framework Convention on Climate Change (UNFCCC), participating countries prepare annual national GHG inventories (NGHGI) that include estimates of anthropogenic GHG emissions by sources and removals by sinks (Environment and Climate Change Canada, 2022; United Nations, 1992). Reliable data in NGHGI are essential for creating effective climate change policies and mitigation strategies, determining compliance with internationally agreed-upon targets, and tracking the sources and trends of GHG emissions and reductions (Gillenwater et al., 2007; Lee et al., 2014, Polish Academy of Sciences, 2015). However, NGHGI data are associated with varying degrees of uncertainty (Fortin, 2021; Ritter et al., 2010).

Uncertainties in the land-use, land-use change, and forestry (LULUCF) sector are larger and more challenging to quantify in comparison to others sectors mainly due to the complexity of the processes involved in the GHG fluxes (Fortin, 2021; McGlynn et al., 2022; Ritter et al., 2010; United Nations, 1992). These uncertainties may result in challenges enacting appropriate mitigation measures, policies, and tracking GHG reduction

progress. This may undermine the confidence of emission reduction claims, particularly for the forestry sector, which is expected to contribute largely to GHG reductions (McGlynn et al., 2022).

The forestry sector is an area of importance that can be used to mitigate climate change by increasing carbon dioxide removals from the atmosphere or halting the loss of carbon stock from the land (UNFCCC, 2020). The more stringent GHG reduction targets rely heavily on land-based mitigation options. Researchers have reported that one-third of GHG reductions needed to prevent the most severe consequences of climate change can be provided by the LULUCF sector (Shukla et al., 2020; UN-REDD Programme, 2022). Above-ground biomass (AGB) is an important carbon pool in forest ecosystems that most carbon estimates are derived from, as the extent of AGB determines current and future carbon storage capacities (Fu et al., 2017). The scope of this research will address uncertainties in large-area carbon quantification from AGB in the forestry sector, as the quantification methods employed on this scale are used to determine estimates for NGHGs. AGB is most commonly quantified using allometric models calibrated from forest inventory data (Návar, 2010). However, measuring the effects of model uncertainties on carbon estimates from AGB is challenging, prompting the need for research related to uncertainty analysis methods (McRoberts and Westfall, 2014; Qin et al., 2021; Wayson et al., 2015).

Uncertainty analysis is an important tool that is required by the IPCC and designated as a good practice to improve the reliability of NGHGI estimates (Lee et al., 2020; Ritter et al., 2010). Uncertainty analysis assesses and documents the causes of uncertainty in individual estimates and the overall total (Paciornik et al., 2019). The information provided by uncertainty analysis can be used to prioritize methodological and data collection improvements (Paciornik et al., 2019). Quantifying uncertainty estimates requires analysts to have a scientific and technical understanding of the carbon fluxes and the quantification methods associated with the system, as uncertainty analysis only provides reliable results if properly implemented (Frey et al., 2006). Hence, there is a need to assess if the IPCC's guidelines for reporting uncertainties in NGHGI's are sufficient when accounting for model uncertainties for complex systems, such as the forestry sector.

1.1.2 Knowledge Gaps

A comprehensive literature review was conducted to: investigate the sources of uncertainty in carbon from AGB quantification and the methods used to quantify these uncertainties, review the IPCC guidelines for uncertainty analysis and identify areas where improvements may be needed, and review studies on uncertainty analysis methods to identify alternative methods that may improve the precision of uncertainty estimates. The scope of the research was established based on the following knowledge gaps identified through the literature review:

1. **Quantifying the Effects of Model Uncertainties:** Carbon emissions and removals from the forestry sector are challenging to quantify. Models calibrated from forest inventory data are commonly used in large-area estimation (Návar, 2010). However, model uncertainty is complex and in practice are rarely quantified (McRoberts and Westfall, 2016; Metsaranta et al., 2017). Based on existing research, there is a need for more studies on quantifying the effects of model uncertainty on carbon estimates from AGB.
2. **Guidance on Implementing Simulation-Based Uncertainty Analysis Methods:** The IPCC recommends using Monte Carlo Simulation (MCS) to quantify uncertainties in complex systems, such as the forestry sector, where large uncertainties are expected (Frey et al., 2006). However, few countries have implemented simulation based uncertainty analysis methods (Monni et al., 2007). By reviewing existing guidelines for MCS, this research suggests that more guidance is needed on how to employ this method when quantifying uncertainties in models involving complex algorithms.
3. **Comparison of Alternative Uncertainty Analysis Methods:** Despite the growing literature on uncertainty analysis methods, guidelines for the use of MCS have not been updated since 2006 (Paciornik et al., 2019). Additional research is needed to assess how alternative uncertainty analysis methods impact uncertainty estimates, and if an alternative method can improve the precision of estimates.
4. **Improved Documentation of Uncertainty Analysis Methods:** Quantifying uncertainty estimates must be comparable between countries for NGHGs to be used for policy purposes and to determine compliance with international

commitments (Gillenwater et al., 2007). Improved documentation can assist countries to better understand and replicate procedures for uncertainty analysis. Additional work is needed to clearly outline the procedure used to quantify model uncertainties per TACCC reporting principles and assess how these methods apply to different countries.

1.1.3 Objectives

This research suggests that improvements can be made in how uncertainty analysis methods are used in NGHGs when quantifying model uncertainties to increase the reliability of uncertainty estimates, improve implementation, and comply with TACCC reporting principles. This research will assess and quantify the uncertainties in carbon estimation from AGB allometric models used in the forestry sector. The uncertainty analysis method proposed can be used by countries to adopt and improve uncertainty quantification, and better understand the effects of uncertainties. The following objectives are proposed in order to achieve the overall objective:

1. Assess the causes and the types of uncertainties associated with carbon quantification from AGB in the forestry sector.
2. Review the IPCC guidelines for conducting uncertainty analysis used in NGHGs. Identify alternative uncertainty analysis methods that may better define uncertainties compared to the methods recommended by the IPCC.
3. Develop a methodology to quantify uncertainties in carbon estimates from AGB.
4. Conduct case studies to investigate how uncertainties can be quantified in carbon estimates from AGB using different uncertainty analysis approaches.
5. Evaluate the impacts of these uncertainties on carbon quantification from AGB.

1.2 Methodology

The objectives are achieved by following the methodology outlined in Figure 1.1.

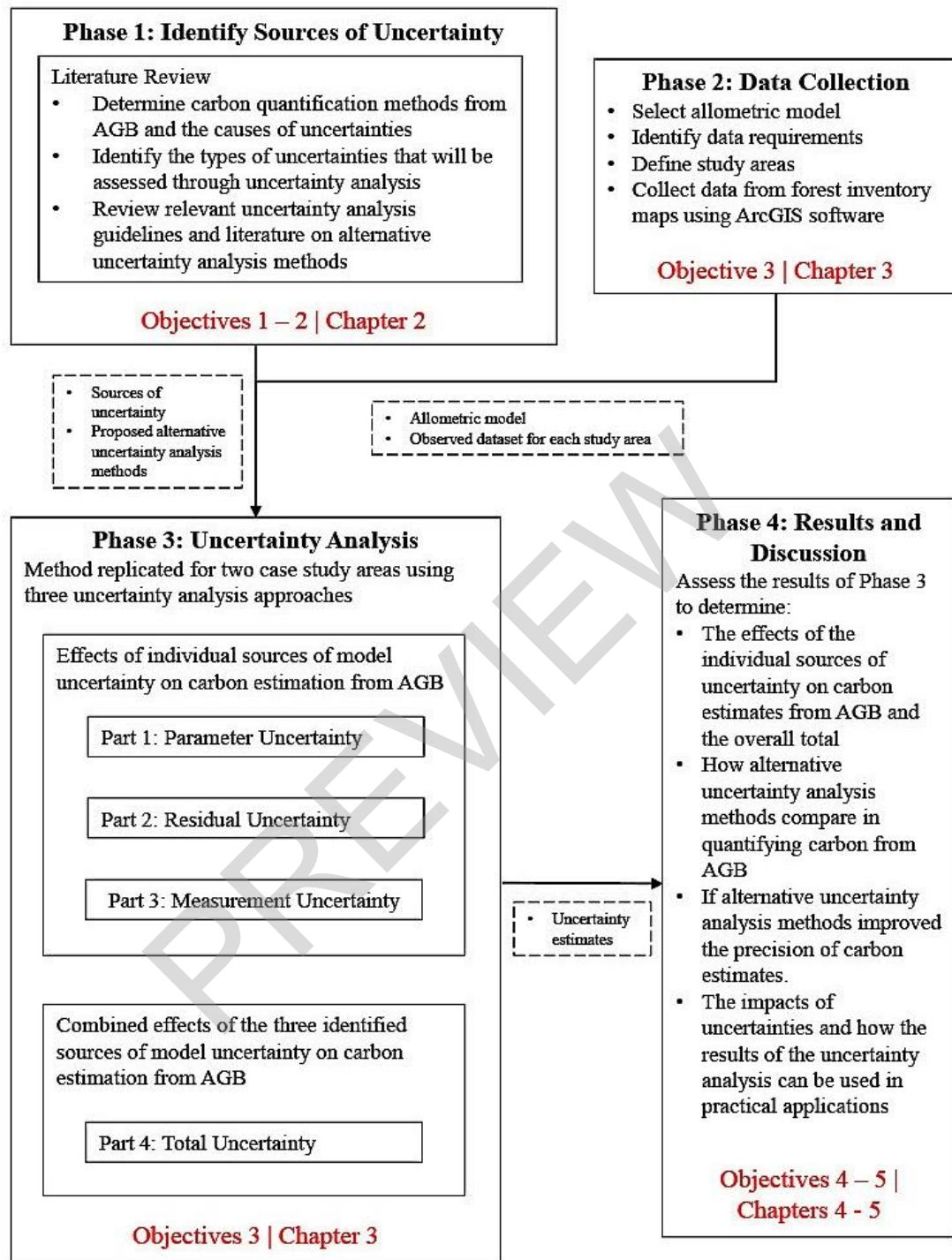


Figure 1.1 Overview of the research methodology

1.2.1 Phase 1: Identifying Uncertainties

Phase 1 consisted of an in-depth literature review which identified the challenges in quantifying carbon from AGB, and the associated sources of uncertainty. For large-area estimates used in NGHGs, the most common method consists of deriving allometric models from forest inventory data that are obtained from direct measurements or remote sensing (Návar, 2010; Picard et al., 2012). The main sources of uncertainty in error analysis for forest inventories are from measurement, sampling, and model uncertainty (Cunia, 1965). Based on the literature, the scope of this study was to determine the causes and quantification of model uncertainty. The sources of model uncertainty are due to variances in model parameters, residual variance, and measurement error in the model input variables (Berger et al., 2014; Fu et al., 2017; McRoberts and Westfall, 2014, 2016; Qin et al., 2021). The uncertainty analysis procedure, including identifying alternative uncertainty analysis approaches, for analyzing uncertainties in carbon from AGB due to model uncertainty was formulated by reviewing guidelines from the IPCC and relevant literature. Phase 1 was concluded by identifying the sources of model uncertainties, and the proposed alternative uncertainty analysis approaches.

1.2.2 Phase 2: Data Collection

Phase 2 consisted of identifying the particular allometric model and the associated data needs. Case studies for an area in British Columbia (B.C.), Canada, and in Västernorrland County, Sweden, were proposed to determine the feasibility of the uncertainty analysis method when used in different countries. Data was collected from forest inventory maps. Phase 2 was concluded by constructing observed datasets for each of the study areas.

1.2.3 Phase 3: Uncertainty Analysis

Phase 3 consisted of a four-part uncertainty analysis. Parts 1 to 3 evaluated each individual source of model uncertainty and their effects on the estimation of carbon from AGB. Part 4 evaluated the combined effects of the sources of model uncertainty to calculate the overall effect on carbon from AGB quantification. The method was repeated for both study areas and for each of the alternative uncertainty analysis approaches identified in Phase 1.

1.2.4 Phase 4: Results and Discussion

Phase 4 consisted of assessing the results of the uncertainty analysis conducted in Phase 3. The results were compared to determine the effects of the individual sources of uncertainty on carbon estimates from AGB, and how results differed for the different study areas. This information was used to recommend methodological and data collection improvements in carbon quantification from AGB, and to assess the impacts of uncertainties on the creation of climate change mitigation strategies and policies. The alternative uncertainty analysis approaches were compared to determine how uncertainty estimates differed and if a method produced more precise estimates. This information was used to determine if recommendations could be made to the IPCC's guidelines for uncertainty analysis.

1.3 Thesis Organization

The thesis consists of five chapters with the following contents:

- Chapter 1. Introduces the background, problem statement, and objectives of the research. The methodology followed to achieve the research objectives presented.
- Chapter 2. Presents a comprehensive literature review that justifies the motivation for the research objectives.
- Chapter 3. Describes the study systems and data collection methodology. An in-depth methodology of the four part uncertainty analysis method is presented.
- Chapter 4. Presents the results of the uncertainty analysis conducted in Chapter 3. The results were interpreted to identify: (1) the main sources of uncertainty, (2) the effects of using alternative uncertainty analysis approaches, (3) the methodological and data collection improvements to reduce uncertainties, and (4) the effects of uncertainties on the creation of climate change mitigation strategies and policies.
- Chapter 5. Summarizes the contributions of the research and discusses the limitations of the study and the opportunities for future work.

2.0 LITERATURE REVIEW

2.1 Introduction

Reliable greenhouse gas (GHG) emission and removal data in national GHG inventories (NGHGs) are essential for planning climate action and policies. This research will investigate the causes and quantification methods for the uncertainties in carbon estimates from above-ground biomass (AGB). The following literature review will:

1. Discuss the need for GHG quantification and monitoring as it relates to current and foreseen challenges to climate change and existing global commitments.
2. Provide a background of GHG sources and sinks from the forestry sector to explain the significance of AGB quantification.
3. Discuss AGB quantification methods and the associated challenges contributing to uncertainties.
4. Identify sources of uncertainties in AGB quantification and review the literature on the methods used to quantify these uncertainties.
5. Review guidelines for uncertainty analysis provided by the Intergovernmental Panel on Climate Change (IPCC) to identify areas where improvements are needed.
6. Review the literature on alternative uncertainty analysis methods to identify methods that will be assessed in this study.

2.2 Overview of Greenhouse Gases and Climate Change

Climate change is one of the greatest challenges affecting many weather and climate extremes in every region across the globe (IPCC, 2021). The IPCC, established by the World Meteorological Organization (WMO) and the United Nations (UN) Environment, aims to provide authoritative and objective scientific and technical sources of information (IPCC, 2014b). According to reports by the IPCC, the scale of recent changes across the climate system as a whole and the present state of many aspects of the climate system are unprecedented over many centuries to many thousands of years (IPCC, 2021). Human influence has unequivocally contributed to atmospheric warming (IPCC, 2021). Anthropogenic GHG emissions have increased significantly since the pre-industrial era (IPCC, 2014b). GHGs absorb heat in the atmosphere and re-release is as infrared radiation back to the Earth's surface (United Nations, 1992). Therefore, mean global

temperature rise increases along with increases in the concentration of GHGs in the Earth's atmosphere (IPCC, 2014b). Climate change is a threat multiplier that will continue to impact human and natural systems, as the severity of adverse impacts is predicted to increase with increases in global temperatures (IPCC, 2018). The IPCC's Sixth Assessment Report found that climate resilient developments are increasingly limited if GHG emissions do not rapidly decline, especially if global warming levels exceeds 1.5°C in the near-term (IPCC, 2022). Developments may not be possible in some regions and sub-regions if global warming levels exceed 2°C (IPCC, 2022). Limiting climate change would require strong and sustained reductions in GHG emissions globally, including a decline of anthropogenic CO₂ emissions by approximately 45% from 2010 levels by 2030, reaching net zero by 2050, while also reducing net non-CO₂ radiative forcing (IPCC, 2018, 2021). This would require rapid and far-reaching transitions, prompting the need for strengthened global responses (IPCC, 2014, 2018).

2.3 A Brief History of Global Responses to Climate Change

The UN first recognized climate change as global concern at The Rio de Janeiro Earth Summit in 1992 (Council on Foreign Relations, 2022). The summit resulted in some of the first international agreements on climate change that would become the foundation for future accords, including the United Nations Framework Convention on Climate Change (UNFCCC) (Council on Foreign Relations, 2022). The UNFCCC acknowledges human activities have substantially increased atmospheric GHG emissions that have caused temperatures to rise in the Earth's surface and atmosphere, adversely affecting natural ecosystems and humankind. (United Nations, 1992). The UNFCCC calls for cooperation between all countries, and their participation in effective and appropriate international responses to climate change, with respect to the capabilities of individual countries (United Nations, 1992). This requires frequent meetings between ratifying countries, leading to the formation of the Conference of Parties (COP) (Council on Foreign Relations, 2022). The ultimate objective of the UNFCCC along with legal agreements adopted by the COP is to stabilize GHG concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system (United Nations, 1992). Strategies to mitigate climate change require accurate GHG emission and removal data that can be used to plan, analyze, validate, and at a global scale verify